SOUMIKA DAS

Software Engineer | Full-Stack & Backend Development | AI/ML

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Education

B.Tech in Computer Science and Engineering (IoT, Cybersecurity and Blockchain Technology) Expected 2027
Institute of Engineering and Management, Kolkata CGPA: 7.69/10

Core Skills

- **Languages:** Python, JavaScript, TypeScript, Java, C++, SQL
- **Frontend:** React, Next.js, HTML, CSS, Tailwind CSS, Vite
- **Backend:** Node.js, Express, FastAPI, Flask, Django, REST APIs, WebSocket, Microservices, JWT/OAuth
- **Databases:** PostgreSQL, MySQL, MS SQL Server, MongoDB, Vector DB
- **Cloud & DevOps:** Google Cloud (GCP), Vercel, Render, Docker, Linux, Nginx, CI/CD, Git/GitHub, Ansible
- **AI/ML:** GenAI, LLMs, RAG, Agentic AI, PyTorch, TensorFlow, OpenCV, NLP, CNN, LSTM, YOLO
- **CS Fundamentals:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Unit Testing

Internship Experience

Software Engineer Intern, PepHub **Oct 2025 - Present** | Remote

- Delivered multiple client projects, including a Next.js/Node.js portal with **Stripe payments** and a FinTech REST API.
- Automated production releases for client apps using PM2 and SSH, cutting deployment time by **82%** (45 min to 8 min).

Research Engineer Intern, IEDC, Dept. of CSE, UEM Kolkata **Dec 2024 - Dec 2025** | Hybrid

- Reduced "Satyameba" HPC cluster setup time by **95%** (8 hrs to 22 min) by automating provisioning with Ansible.
- Implemented real-time YOLO/OpenCV face detection at **96% accuracy** and 45 FPS across 8 concurrent video streams.

Software Engineer Intern, Cardiacrhythm **Jun 2025 - Sep 2025** | Remote

- Architected real-time WebSocket ECG streaming for **200+ live streams**, cutting latency by **90%** (1.8s to 180ms).
- Shrank payloads by **75%** (8MB to 2MB) and added auto-reconnect, preventing data loss in **15K+ daily sessions**.

Software Engineer Intern, RevMaxx **Jan 2025 - Jun 2025** | Onsite

- Built a **HIPAA-compliant** transcription backend with SNOMED CT/ICD-10 coding and NLP across **50K+ clinical notes**.
- Added retry logic and circuit breakers to backend services, cutting service failures by **over 90%** (12/week to under 1).

Projects

BONAP - Botanical Scraper & Plant Portal | Live App

- Developed a Node.js scraper and ETL pipeline pulling taxonomy, distribution maps and morphology from BONAP, USDA and GBIF into a Vite + React comparison portal covering **26K+ species** across 4K+ genera, with AI summaries.

Brain Tumor Classification - Ensemble Deep Learning

- Engineered a CNN-LSTM ensemble classifying brain tumors from **3,200+ MRI scans**, reducing single-model variance.

DeepTrust - Degree Verification API | GitHub

- Designed a Flask API verifying degrees via QR codes and OpenCV/Tesseract OCR, supporting PDFs and **6 image formats**.

Publications & Leadership

- **Enhancing Brain Tumor Diagnosis with Ensemble Deep Learning Technique for Improved Accuracy**, AISC 2025 (Springer) | LinkedIn
- **Cloud Lead & Speaker**, GDG on Campus UEM Kolkata (2025 - 26): led the Cloud domain, conducted Google Cloud Study Jams, spoke at "Tech and Adda" and co-organized the "StackUp!!" workshop. | LinkedIn